



GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Machine Learning

DS-552-MJ

Sample / Practice Question Paper

Semester II · Core · Practice Paper

& This is a SAMPLE / PRACTICE question paper prepared for revision from the SPPU M.Sc. Data Science syllabus. It is NOT an official or past university examination paper. Use it only to test your preparation.

Time: 3 Hours

Maximum Marks: 70

Instructions: (1) All questions carry equal weightage where indicated. (2) Figures to the right indicate full marks. (3) Draw neat diagrams / write code wherever necessary.

Q1. Answer any FIVE of the following. [5 × 2 = 10]

- a) Differentiate supervised and unsupervised learning.
- b) What is overfitting?
- c) Define the bias-variance tradeoff.
- d) What is regularization?
- e) State precision and recall.
- f) What is a confusion matrix?
- g) What is the K-Means algorithm?

Q2. Answer the following. [15]

A) Explain linear, multiple and polynomial regression with residual analysis.
OR

B) Explain regularization methods: Lasso, Ridge and ElasticNet.

Q3. Answer the following. [15]

A) Explain SVM, KNN and Naïve Bayes classification algorithms.
OR

B) Explain decision tree classification (CART) and model evaluation using ROC.

Q4. Answer the following. [15]

A) Explain clustering methods: K-Means, hierarchical and density-based.
OR

B) Explain association rule mining: Apriori, Eclat and FP-trees.

